

115. The standard deviation of 100 observations is 10. If 5 is multiplied to each of the observations, then what is the new standard deviation ?

- (a) 20
- (b) 25
- (c) 40
- (d) 50

116. For a Binomial distribution with mean 6 and standard deviation $\sqrt{2}$, what is the value of $P(X = 0)$?

- (a) $\left(\frac{1}{3}\right)^9$
- (b) $\left(\frac{2}{3}\right)^9$
- (c) $\frac{1}{3}\left(\frac{2}{3}\right)^8$
- (d) $\frac{2}{3}\left(\frac{1}{3}\right)^8$

117. If the random variable X has mean 3 and standard deviation 5, then what is the variance of the random variable $Y = 2X - 5$?

- (a) 15
- (b) 40
- (c) 45
- (d) 100

118. Three events A , B and C are such that A and B are disjoint, A and C are independent, B and C are independent. If $4P(A) = 2P(B) = P(C)$ and $P(A \cup B \cup C) = 5P(A)$, then what is the value of $P(C)$?

- (a) $\frac{5}{6}$
- (b) $\frac{1}{3}$
- (c) $\frac{1}{6}$
- (d) $\frac{2}{3}$

For the next **two (02)** items that follow :

The frequency distribution of marks obtained by 100 students in a certain subject is given below :

Marks	0-10	10-20	20-30	30-40
No. of students	10	20	f	40

119. What is the mean deviation about the arithmetic mean ?

- (a) 5.5
- (b) 6
- (c) 6.5
- (d) 8

120. What is the standard deviation ?

- (a) 8.5
- (b) 9
- (c) 9.5
- (d) 10

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